

Article

A Comparative Study on Battery Modelling via Specific Hybrid Pulse Power Characterization Testing for Unmanned Aerial Vehicles in Real Flight Conditions

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Abstract: Battery modelling is essential for optimizing the performance and reliability of Unmanned Aerial Vehicles (UAVs), particularly given the challenges posed by their dynamic power demands and limited onboard computational resources. This study evaluates two widely adopted Equivalent Circuit Models (ECMs), the fixed resistance model and the Thevenin model to determine their suitability for UAV applications. Using the Specific Hybrid Pulse Power Characterization (SHPPC) method, key parameters, including Open Circuit Voltage (OCV), internal resistance (R_i), polarization resistance (R_1), and polarization capacitance (C_1), were estimated across multiple states of charge (SOC). The models were analyzed under nine parameterization scenarios, ranging from fully average parameters to configurations where selected parameters were tied to SOC. Results indicate that the Thevenin model, with selective SOC-dependent parameters, demonstrated superior predictive accuracy, achieving error reductions of up to 4.26 times compared to the fixed resistance model. Additionally, findings reveal that modelling all parameters as SOC-dependent is unnecessary, as simpler configurations can balance accuracy and computational efficiency, particularly for UAVs with constrained BMS capabilities.

Keywords: UAV; battery parameter estimation; SHPPC; Equivalent Circuit Model; real flight data



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1. Introduction

The Unmanned Aerial Vehicle (UAV) industry is continuously growing [1,2], with broad applications in both military and civil sectors. UAVs perform diverse missions, including operations in inaccessible or hazardous areas [3–5], and play critical roles in traffic monitoring, highway inspection, surveying, agriculture, and urban logistics. These expanding applications drive ongoing technological development and integration into UAV systems, where energy efficiency and reliability remain crucial [6–8].

A vital component of UAV systems is the battery, which serves as the primary power source for electrically driven UAVs. Electric propulsion offers several advantages over fossil fuel-powered systems, including greater flexibility in power options, lower operational costs, easier maintenance, and environmental benefits. However, the relatively short flight

time of battery-powered UAVs, caused by the lower energy density of batteries compared to fossil fuels, poses a significant challenge [9–11].

Battery modelling is essential for understanding battery behaviour, optimizing performance, and ensuring safe operation, particularly under dynamic load conditions [3–5,12]. UAVs often experience rapid and unpredictable power demand changes during critical phases like take-off, hovering, and manoeuvring. These demands necessitate real-time battery status estimation and predictive modelling to maintain operational reliability [13–17]. The nomenclature used throughout this study is presented in Table 1 for clarity and ease of reference.

Table 1. Nomenclature.

Abbreviation	Definition
OCV	Open Circuit Voltage
Ri	Internal Resistance of the battery
R1	Polarization Resistance in the battery circuit
C1	Polarization Capacitance
SOC	State of Charge, percentage of the remaining capacity
SHPPC	Specific Hybrid Pulse Power Characterization method
UAV	Unmanned Aerial Vehicle
BMS	Battery Management System
ECM	Equivalent Circuit Model
n	Order of the Equivalent Circuit Model (e.g., n-RC model)
Ah	Battery capacity in Ampere-Hours
TOW	Take-off Weight
RC	resistor–capacitor networks
n-RC	Higher-order ECM (e.g., 2-RC, 3-RC depending on the number of RC)
SHPPC_nx_ML	Parameters estimation using SHPPC with mean load
SHPPC_nx_PL	Parameters estimation using SHPPC with peak load
OCV_PL_nx	OCV estimation of the n = x model using SHPPC with peak load
Ri_nx_ML (PL)	Ri estimation using SHPPC with mean load (peak load)
Ri_avg_nx	Mean value of Ri estimation using SHPPC
Vt	Measured battery voltage
s_k	Voltage predicted by the battery model

Several studies have demonstrated the feasibility of using constant resistance battery models for UAV applications, as they balance simplicity and efficiency [5,18–22]. For example, in hybrid UAV systems, constant resistance models are often integrated into energy management strategies to optimize power distribution between batteries and fuel cells. These models are computationally efficient and well-suited for UAVs with constrained on-board processing capabilities, as highlighted by Boukoberine et al. [18], who implemented a rule-based energy management strategy using a constant resistance battery model to enhance UAV endurance and hydrogen consumption efficiency. Battery models are generally categorized into three types: Electrochemical Models (EMs), Black Box Models, and Equivalent Circuit Models (ECMs). EMs simulate internal chemical reactions with high accuracy but require significant computational resources, making them unsuitable for UAVs. Black Box Models, including machine learning-based approaches, depend heavily on extensive training data and often fail to generalize beyond the training conditions. ECMs, by contrast, offer a practical balance of simplicity and functionality by approximating battery behaviour using electrical circuit components, making them particularly suitable for UAV applications [2,5,23,24].

A critical aspect of battery modelling is parameterization. While parameters such as Open Circuit Voltage (OCV), internal resistance (Ri), and polarization characteristics (R1, C1) are inherently related to the battery's state of charge (SOC), accounting for all these

dependencies can result in overly complex models. Simplifying certain parameters, for instance by averaging, can significantly reduce computational demands without substantial loss of accuracy for many applications [6,25,26]. The choice of which parameters vary must therefore balance model complexity with predictive accuracy, depending on the intended application.

Additionally, increasing the order of ECMs, such as by incorporating more resistor–capacitor (RC) networks, does not always result in proportional gains in accuracy. Studies have shown that while higher-order models can capture finer battery dynamics, the marginal improvements in accuracy often fail to justify the additional computational burden [25–28]. This trade-off is particularly relevant for UAVs, where processing resources are limited, and efficient models are critical for real-time operation.

Despite significant advancements in battery modelling for electric vehicles (EVs), UAV-specific applications remain underexplored. UAVs face unique challenges, such as rapid power surges and fluctuating load demands, which require tailored modelling approaches. Addressing these gaps is essential for optimizing UAV performance in real-world missions.

To validate the proposed battery models under real-world conditions, this study employed a multirotor UAV configured for surveying and mapping missions, as shown in Figure 1. The UAV operated across three distinct mission phases: take-off, surveying, and landing with varying speed, altitude, and load current demands. The specifications of the UAV and its mission profile are summarized in Table 2.

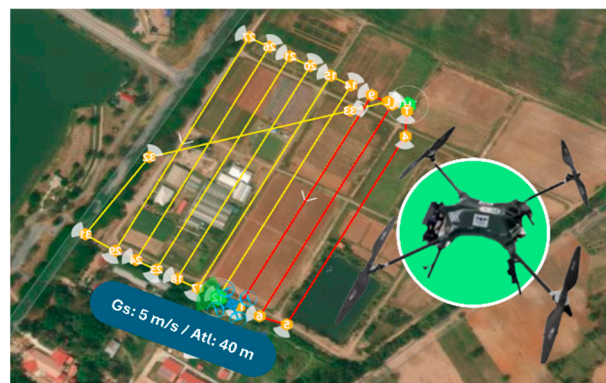


Figure 1. Multirotor UAVs and Flight Trajectory for Surveying and Mapping.

Table 2. UAV Specifications and Mission Profile.

Parameters	Description
Propulsion	Electrical Motors
TOW	7.5 kg
Payload	2.5 kg (Batt. 0.5 kg + Camera 1.5 kg)
Source	Li-Po battery pack, 16 Ah-6S-22.2 V
Profiles	Speed, Altitude, Load Current
Take-off	5 m/s, 20 m, 65 A
Survey	5 m/s, 40 m, 31 A
Landing	5 m/s, 20 m, 20 A

This study evaluates two commonly used ECMs, the fixed resistance model (n-RC model, $n = 0$) and the Thevenin model (n-RC model, $n = 1$), to determine their suitability for UAV applications. The Specific Hybrid Pulse Power Characterization (SHPPC) method, a reliable approach for parameter estimation, is employed to estimate key parameters such as Open Circuit Voltage (OCV), internal resistance (R_i), polarization resistance (R_1), and polarization capacitance (C_1) across multiple SOC levels [29–34]. These parameters are

known to vary significantly with SOC or capacity (Ah), influencing the battery's dynamic response. For instance, OCV decreases as the battery discharges, while R_i , R_1 , and C_1 exhibit nonlinear relationships with SOC, particularly under high power demands.

However, modelling all parameters as functions of SOC or Ah can lead to increased model complexity [24,35], which poses significant challenges for real-time applications. UAVs, unlike electric vehicles (EVs), rely on lightweight and low-power Battery Management Systems (BMS) to minimize overall weight and maximize operational efficiency. These BMS units often have limited computational capabilities, making it impractical to adopt models with excessively complex parameterizations. Therefore, this study aims to strike a balance between accuracy and computational feasibility by exploring scenarios where only select parameters are treated as SOC-dependent, while others are held constant.

The models are analyzed under nine parameterization scenarios, ranging from fully average parameters (simplified constant values) to configurations where specific parameters are tied to SOC. This systematic approach provides insights into the trade-offs between predictive accuracy and computational simplicity. Initial results demonstrate that the Thevenin model ($n = 1$), with selective SOC-dependent parameters, achieves superior predictive performance compared to the fixed resistance model ($n = 0$), while remaining computationally efficient. By exploring which parameters should vary with SOC to retain accuracy while minimizing computational load, this study highlights the potential to develop simplified and reliable battery models tailored to the specific needs of UAV operations. The findings are particularly significant given the constraints of UAV systems, which prioritize lightweight and power-efficient designs over high computational capacity. These results underscore the importance of strategic parameterization in ECMs, offering practical guidelines for future UAV BMS development.

This paper is organized as follows: Section 2 introduces battery models and parameterization methods. Section 3 describes the experimental setup and model evaluation. Section 4 presents the results, and discusses the findings and concludes this study.

2. Materials and Methods

2.1. Battery Model

Equivalent Circuit Models (ECMs), as shown in Figure 2, were employed due to their ability to represent the physical characteristics of the battery through their parameters [34]. This approach not only provides meaningful insights into the battery's behaviour but also facilitates future analysis to explore its relationships with other influencing factors [24]. Analysing the n-RC Equivalent Circuit Model shown in Figure 2 using Kirchhoff's Current Law yields the following:

$$V_t(s) = V_{oc}(s) - i_L(s) \left(R_i + \frac{R_1}{1 + R_1 C_1 s} + \dots + \frac{R_n}{1 + R_n C_n s} \right) \quad (1)$$

when $n \in \mathbb{N}_0$, $V_t(s)$ refers to the terminal voltage, $V_{oc}(s)$ corresponds to the Open Circuit Voltage (OCV), and $i_L(s)$ represents the load current at the k th sampling interval. After discretizing Equation (1), we derive the following expression [24,34].

$$V_{t,k} = \left(1 - \sum_{i=1}^n g_i \right) V_{oc,k} + g_1 V_{t,k-1} + g_2 V_{t,k-2} + \dots + g_n V_{t,k-n} + g_{n+1} i_{L,k} + g_{n+2} i_{L,k-1} + \dots + g_{2n+1} i_{L,k-n} \quad (2)$$

where $i = 1, 2, \dots, 2n + 1$

$$\Psi_{n,k} = [1 V_{t,k-1,2} \dots V_{t,k-n} i_{L,k} i_{L,k-1,2} \dots i_{L,k-n}] \quad (3)$$

$$\Theta_{n,k} = \left[\left(1 - \sum_{i=1}^n c_i \right) V_{oc,k} \ g_1 \ g_2 \ g_3 \ \dots \ g_{2n+1} \right]^T \quad (4)$$

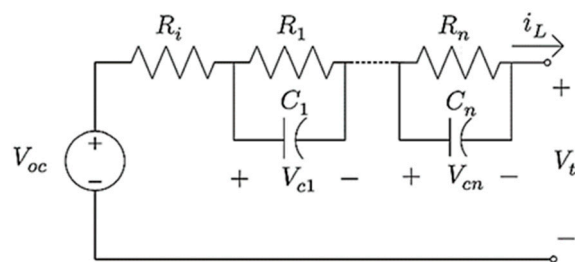


Figure 2. n-RC Equivalent Circuit Models.

The equation representing the model can be formulated as

$$s_k = \Psi_{n,k} \Theta_{n,k} \quad (5)$$

where $s_k = V_{t,k}$ from Equation (5), battery parameters identification can be performed as follows:

$$\Theta_{n,k} = \Psi_{n,k}(s_k)^{-1} \quad (6)$$

In this study, we utilized Equation (5) to predict the terminal voltage of the battery under varying load conditions. This equation models the dynamic behaviour of the battery by incorporating key parameters. Equation (6) was employed to estimate these parameters, providing a systematic approach to characterize the battery's performance across different states of charge (SOC). These equations form the basis for evaluating the battery models under various parameterization scenarios, enabling insights into the trade-offs between accuracy and computational efficiency.

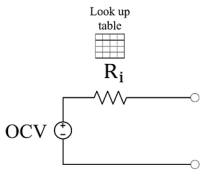
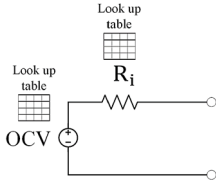
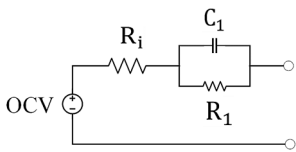
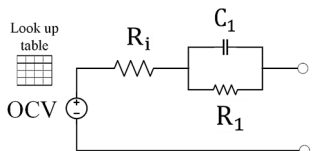
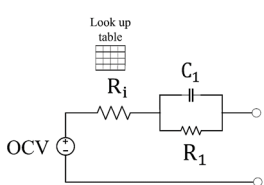
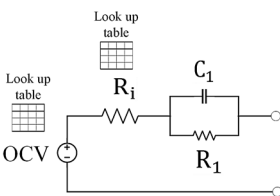
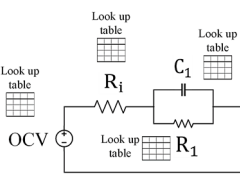
2.2. Overview of Parameter Estimation Test and Models Comparison

This research was the study of influence of parameter estimation from various battery models using the SHPPC test for Unhuman Aerial Vehicle (UAV) with the authentic UAV loading profile test. Initially, this study began with a Specific Hybrid Pulse Power Characterization (SHPPC) battery test. The result obtained was calculated to find Open Circuit Voltage (OCV) followed by estimation of R_i , R_1 , and C_1 from Equation (6). The next step was the creation of 9 battery models. Models 1.0–1.3 were the n-RC which $n = 0$. Models 2.0–2.4 were n-RC which $n = 1$. Details of the models can be found in Table 3. Batteries were brought to the discharge test using the UAV loading profile test. Battery current and voltage behaviours were compared by measured value and prediction from each model. The final step was analysis and conclusion of test results.

Table 3. The different types and conditions of battery models under SHPPC testing.

Model Name	Symbols	Description
Model 1.0		n-RC Model which $n = 0$. All parameters were means and calculated from the mean value from the test (OCV and R_i were means).
Model 1.1		n-RC Model which $n = 0$. OCV value related to capacity when R_i was means.

Table 3. Cont.

Model Name	Symbols	Description
Model 1.2		n-RC Model which $n = 0$. OCV was means and calculated from the mean value when R_i related to capacity.
Model 1.3		n-RC Model which $n = 0$. OCV and R_i related to capacity.
Model 2.0		Thevenin's model or n-RC Model which $n = 1$. OCV, R_i , R_1 , and C_1 were means.
Model 2.1		n-RC Model which $n = 1$. OCV related to capacity while R_i , R_1 , and C_1 were means.
Model 2.2		n-RC Model which $n = 1$. R_i was related to capacity while OCV, R_1 , and C_1 were means.
Model 2.3		n-RC Model which $n = 1$. OCV and R_i related to capacity while R_1 and C_1 were means.
Model 2.4		n-RC Model which $n = 1$. All parameters related to capacity.

The battery test for parameter estimation to create models in this research employed SHPPC. Batteries discharged pulses as equal as mean current and peak current from the authentic loading profile. Hence, this test revealed battery behaviour responding to real loading profile and Open Circuit Voltage. Steps of the tests were as shown in Figure 3.

The testing procedure in SHPPC Battery Testing:

1. Initially, the fully charged battery discharged at 31 A which was the authentic mean load current for 10 s.
2. Discharge stopped for 30 s.
3. Authentic peak discharge was at 65 A for 5 s.
4. Discharge stopped for 30 s.
5. Step 1–4 were repeated until discharge reached 14.60 Ah which was 90% SOC.

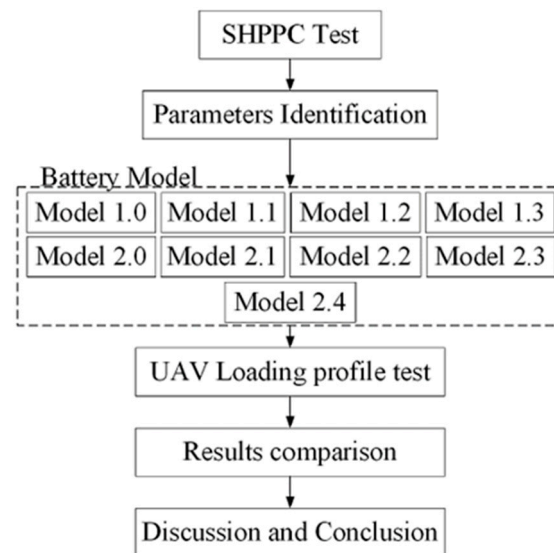


Figure 3. Testing flow chart.

The test used 1-s sampling time [9,24]. The test used an IT6000C Bi-directional Programmable DC Power Supply, as shown in Figure 4, to assign a discharge profile according to the authentic UAV loading profile. The Open Circuit Voltage, current, and electric voltages responding to the authentic loading profile in all assigned capacities were obtained, as presented in Figure 5. The authentic UAV loading profile is illustrated in Figures 6 and 7. The SOC in this research was less than 2%. The test was ongoing for 3 cycles until the batteries ran out of capacity. From this step, the remaining parameter can be estimated using Equation (6). This research was the study of influence of parameter estimation from various battery models.

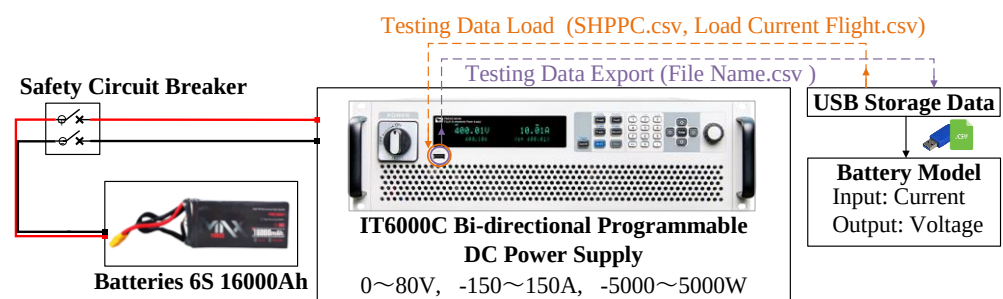


Figure 4. Battery test bench.

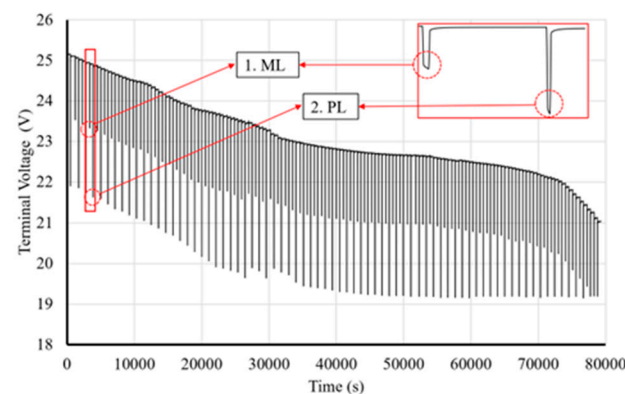


Figure 5. SHPPC Battery test. (The red arrow indicates ML, which represents the voltage response to the Mean Load, and PL, which represents the voltage response to the Peak Load).

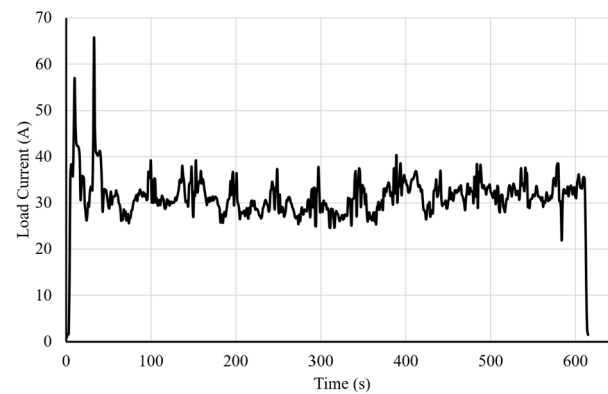


Figure 6. Load current profile of UAV flight in 1 cycle.

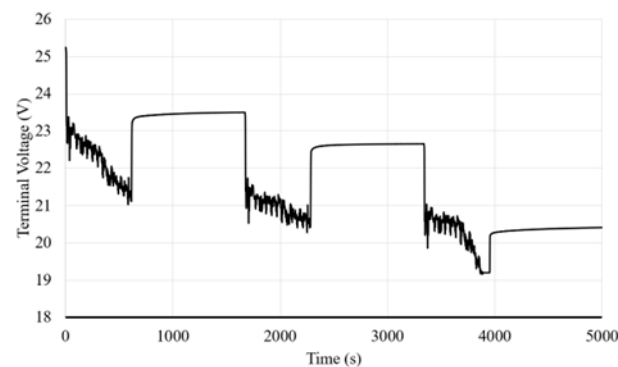


Figure 7. Voltage profile of UAV flight in 3 cycles.

3. Results and Discussion

The test results consist of 3 parts. The first part was the parameters estimation of each model. The second part was the results of calculation or estimation of the electric voltage of each model compared with the battery terminal voltage measurement which used the previously estimated parameters. The last part was the comparison of the Sum Squared Error (SSE) and the Mean Squared Error (MSE) of each model, respectively.

3.1. The Parameters Estimation

Figures 8 and 9 show the estimated parameters of internal resistance or R_i of n -RC Model which $n = 0$. This was the so called fixed resistance model. The model which $n = 1$ was so known as the Thevenin Equivalent Circuit Model or, in short, the Thevenin Model. In both figures, the solid line represented R_i estimated from the fixed resistance model and the Thevenin model using the SHPPC test with mean load. Dots represented the estimation of R_i using SHPPC with peak load. The dashed line was the mean value of the model using the SHPPC test with mean load. This mean value was chosen because the mean load was the load for which the battery most supplied the power. Considering the quality of R_i of both models, slightly similar changes according to capacity were found at no more than just 9%. The parameters estimated by SHPPC with peak load presented a value approximate to the parameter estimated from test with mean load. Internal resistance shown in Figures 8 and 9 was the potential to increase when supplying load. Until load supply reached 5 Ah or SOC at 70%, the resistance lowered and was stable. It was somehow possible to increase the resistance again when the battery was close to fully discharged.

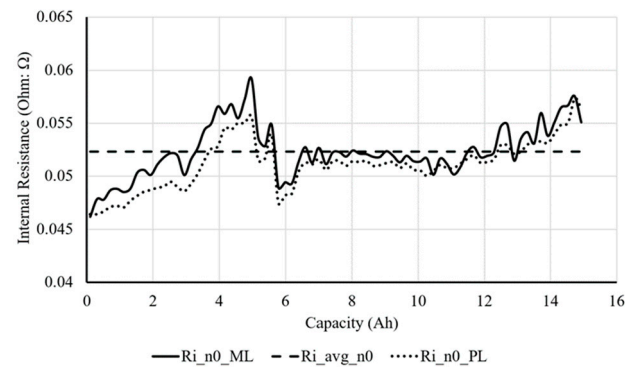


Figure 8. Relation between capacity and internal resistance of the fixed resistance model.

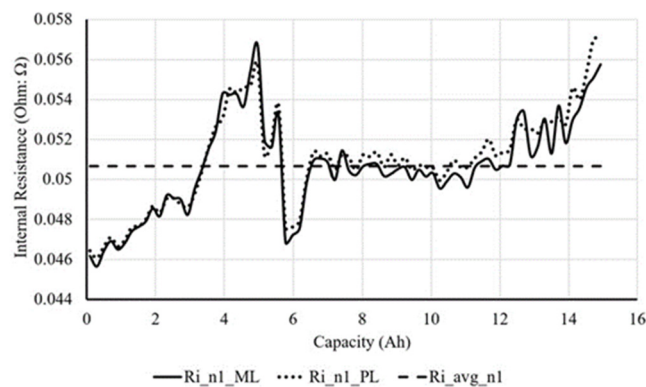


Figure 9. Relation between capacity and internal resistance of the Thevenin model.

In Figure 10, relations between the capacity and polarization resistance or $R1$ of the Thevenin model when both mean load and peak load were fluctuating showed that the parameters estimated from the peak load were potentially slightly lower than the mean load and were approximate to the mean value of the parameter estimated using SHPPC with the mean load. At the end of the test, when the battery was nearly fully discharged, the resistance increased.

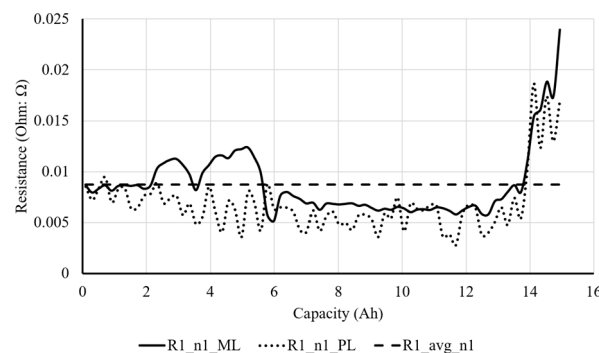


Figure 10. Relation between capacity and polarization resistance ($R1$) of the Thevenin model.

The polarization capacitor estimated from the Thevenin model as shown in Figure 11 obtained by the peak load test (SHPPC_n1_PL) was more fluctuating than that obtained by the mean load (SHPPC_n1_ML). It was also approximate to the mean of resistance tested by SHPPC_n1_ML. Capacitor tended to lower when reaching the fully discharged stage or when the load was discharged to 13.8 Ah or SOC was 13.75%.

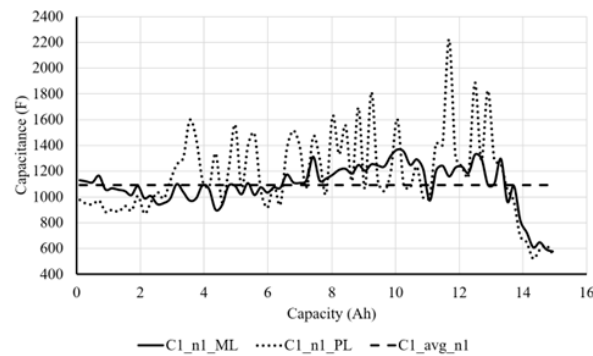


Figure 11. Relation between capacity and the polarization capacitor (C1) of the Thevenin model.

Open Circuit Voltage (OCV) estimation obtained from the fixed resistance model and the Thevenin model are shown in Figure 12. When comparing the measured value of OCV_ML of both models using the SHPPC test with mean load and peak load, their OCV estimations were approximate and tended to be similar to the measured value. There was an exception for estimation of the fixed resistance model using SHPPC with peak load (OCV_PL_n0). Apparently, it presented a lower OCV than the other models.

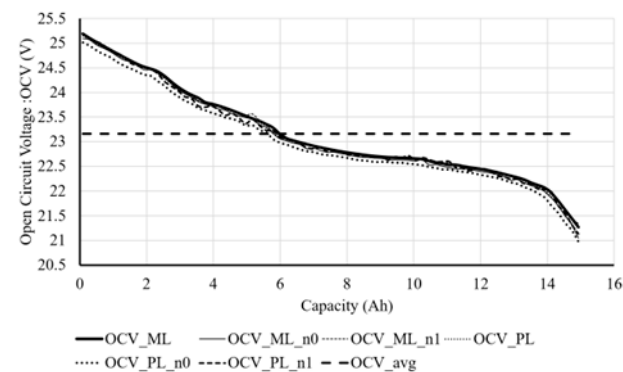


Figure 12. Relation between capacity and OCV of each model.

3.2. Battery Modelling and Comparison

Figures 13 and 14 compare battery terminal voltage from the measured value and the estimation of the battery terminal voltage of each model. From Figure 13, it is shown that Model 1.1 and 1.3 shared a similar pattern of voltage with the measured value, compared to other fixed resistance models.

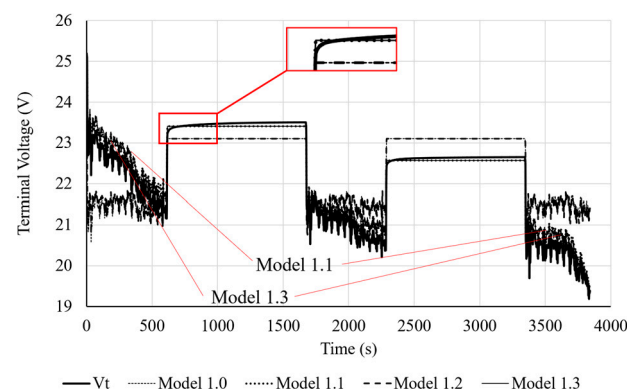


Figure 13. Terminal voltage comparison of the fixed resistance model.

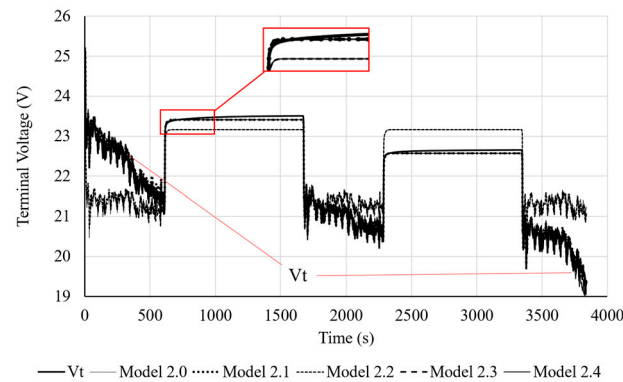


Figure 14. Terminal voltage comparison of the Thevenin model.

Figure 14 shows the comparison for the Thevenin model according to several conditions. Table 3 separates Models 2.0–2.4 and revealed that the models that presented a similar estimation of battery voltage to the measured value were Models 2.1, 2.3, and 2.4. Considering the behaviour of voltage when there was no electric current or polarization characteristics, it was found that the Thevenin model can imitate such behaviour better than the fixed resistance models.

Table 4 presents a comparison of the computational time and parameter characteristics for different battery models. Models with fewer SOC-dependent parameters demonstrate lower computational time, whereas Model 2.4, with all parameters being SOC-dependent, requires the longest time (1.2880 s).

Table 4. Battery terminal voltage estimation results.

Model Name	Total Parameters in the Model	Number of SOC-Dependent Parameters	Estimation Time (s)
Model 1.0	2	0	0.0244
Model 1.1	2	1	0.3370
Model 1.2	2	1	0.3535
Model 1.3	2	2	0.5758
Model 2.0	4	0	0.0282
Model 2.1	4	1	0.3580
Model 2.2	4	1	0.3541
Model 2.3	4	2	0.5924
Model 2.4	4	4	1.2880

3.3. The Comparison of Error

Figures 15 and 16 present the comparison of the Sum Squared of Error (SSE) of the fixed resistance models and the Thevenin models. Figure 15 shows that the load supply was divided into 3 flights. The models with the lowest SSE were Models 1.1 and 1.3. Comparing errors between the two models, it showed that Model 1.1 presented more stable errors than Model 1.3. Model 1.3 presented increasing errors at the beginning of the test which was the moment that the UAV rapidly discharge load took off and SSE was decreasing more than Model 1.1 at the end of the test.

Regarding the SSE of models 1.0 and 1.2, which were the models that did not have a relation between OCV and capacity, the mean value was applied to these models and, apparently, it was more than for Models 1.1 and 1.3. The SSE of each flight was varied which showed that the battery parameters related to changes in capacity.

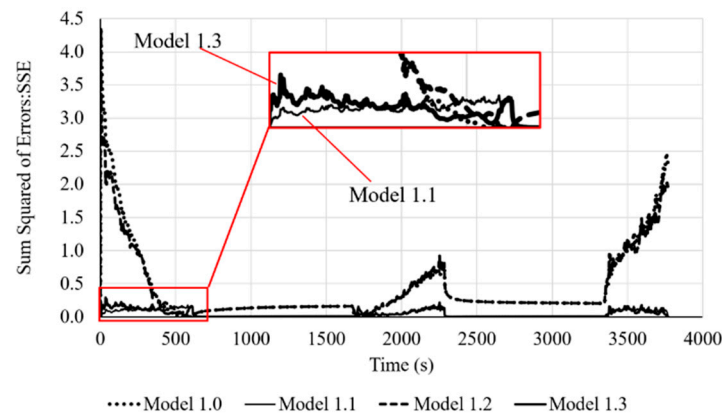


Figure 15. SSE comparison of the fixed resistance model.

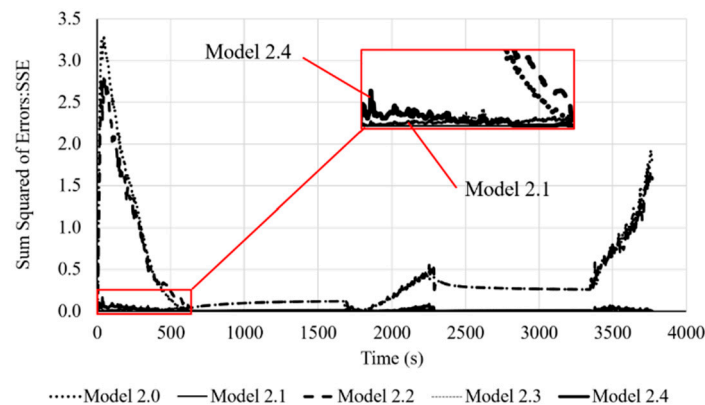


Figure 16. SSE comparison of the Thevenin mode.

Figure 16 shows the SSE of the Thevenin model in each condition, and it revealed that Models 2.1, 2.3, and 2.4 clearly had a lower SSE than other models which used OCV as the means. This complied with Figure 15 showing that Models 2.3 and 2.4 shared similar behaviour. Their SSE were more than Model 2.1 at the beginning of the test and lower at the end. Figure 17 shows a comparison of the Mean Squared Error (MSE) of each model. The model with the least errors was the Thevenin model which were Models 2.1, 2.3, and 2.4, respectively. Fixed resistance models that showed the least errors were Models 1.1 and 1.3. All of them used OCV that related to changes in capacity. Meanwhile, all models with OCV as means showed high errors.

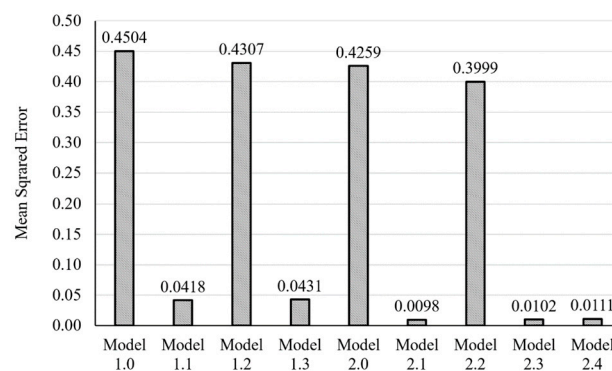


Figure 17. MSE comparison of each model.

4. Conclusions

This section presents the results of the parameter estimation and model test with the authentic load profile of UAV for surveying and mapping. Each model was compared by

different conditions of parameters to find the least complex models that were still accurate to predict battery behaviour. The results can be concluded as follows.

The Thevenin models presented better polarization characteristics than the fixed resistance models. By the SHPPC test using mean load current discharge at 31 A or 1.94 C or 1.94 time of capacity and peak current at 65 A or 4.06 C, it was found that the parameter estimations were approximate.

Models using OCV relating to changes in capacity showed less errors than those using OCV as means. It showed that changes in OCV to capacity clearly affected the accuracy of the model.

Choosing to use R_0 as a means obtained from the mean value of R_0 that was related to changes in capacity did not influence the accuracy of the model when using batteries in the mentioned manner. Since changes in resistance were not high and were 9% from the mean, it was possible to substitute the relation by means. Even though R_1 and C_1 were related to changes in battery capacity, this did fluctuate, and in general, was approximate to means, so it can be substituted by means as well. As mentioned, it can be concluded that the parameters of R_0 , R_1 , and C_1 did not significantly relate to capacity that influenced the models' accuracy.

Among the tested models, those with fewer SOC-dependent parameters achieve faster computational time while maintaining reasonable accuracy. Specifically, Model 2.1 strikes an optimal balance, offering high accuracy with lower computational time compared to more complex models like Models 2.3 and 2.4. This finding, supported by Figure 17, highlights Model 2.1 as a suitable choice for UAV applications where both computational efficiency and accuracy are critical.

To bring the model to work, Model 1.1 was suitable when accuracy was not strictly required and was easier to calculate what was demanded, both for the parameter estimation and for the brief voltage estimation by the model. When a model presenting more accurate battery behaviour was demanded, Model 2.1 was accurate and less complex than Models 2.3 and 2.4. Hence, it was suitable as UAV battery model for surveying and mapping.

In summary, this study demonstrates that by carefully selecting SOC-dependent parameters, it is possible to achieve a balance between model accuracy and computational simplicity, which is critical for UAV applications with constrained processing capabilities. The findings provide practical insights into optimizing battery models, paving the way for further advancements in UAV energy management.

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References

1. Commercial Drone Market Size, Share & Trends Analysis Report by Product (Fixed-Wing, Rotary Blade, Hybrid), by Application, by End-Use, by Region, and Segment Forecasts, 2023–2030. Available online: <https://www.grandviewresearch.com/industry-analysis/global-commercial-drones-market> (accessed on 28 October 2023).

2. Perreault, M.; Behdinan, K. Delivery drone driving cycle. *IEEE Trans. Veh. Technol.* **2021**, *70*, 1146–1156. [[CrossRef](#)]
3. He, Z.; Gómez, D.M.; Hueso, A.d.I.E.; Peña, P.F.; Lu, X.; Moreno, J.M.A. Battery SOC Estimation for Hybrid-Power UAVs Using Fast-OCV Curve with Unscented Kalman Filters. *Sensors* **2023**, *23*, 6429. [[CrossRef](#)] [[PubMed](#)]
4. Saha, B.; Quach, C.C.; Goebel, K. Optimizing battery life for electric UAVs using a Bayesian framework. In Proceedings of the 2012 IEEE Aerospace Conference, Big Sky, MT, USA, 3–10 March 2012.
5. Jiao, S.; Zhang, G.; Zhou, M.; Li, G. A Comprehensive Review of Research Hotspots on Battery Management Systems for UAVs. *IEEE Access* **2023**, *11*, 84636–84650. [[CrossRef](#)]
6. Di Rito, G.; Suti, A.; Ricci, A.; Galatolo, R.; Mattei, G. Experimental characterisation of Li-Po battery packs and BLDC machines for hybrid propulsion systems of lightweight UAVs. In Proceedings of the 2022 IEEE 9th International Workshop on Metrology for AeroSpace (MetroAeroSpace), Pisa, Italy, 27–29 June 2022.
7. Yan, H.; Yang, S.-H.; Chen, Y.; Fahmy, S.A. Optimum Battery Weight for Maximizing Available Energy in UAV-Enabled Wireless Communications. *IEEE Wirel. Commun. Lett.* **2021**, *10*, 1410–1413. [[CrossRef](#)]
8. Chittoor, P.K.; Chokkalingam, B.; Mihet-Popa, L. A Review on UAV Wireless Charging: Fundamentals, Applications, Charging Techniques and Standards. *IEEE Access* **2021**, *9*, 69235–69266. [[CrossRef](#)]
9. Andrea, D. *Battery Management Systems for Large Lithium-Ion Battery Packs*; Artech House: Norwood, MA, USA, 2010.
10. Savkin, A.V.; Huang, H. Deployment of unmanned aerial vehicle base stations for optimal quality of coverage. *IEEE Wirel. Commun. Lett.* **2018**, *8*, 321–324. [[CrossRef](#)]
11. Pan, Y.; Chen, Q.; Zhang, N.; Li, Z.; Zhu, T.; Han, Q. Extending Delivery Range and Decelerating Battery Aging of Logistics UAVs Using Public Buses. *IEEE Trans. Mob. Comput.* **2023**, *22*, 5280–5295. [[CrossRef](#)]
12. Chang, J.; Wei, Z.; He, H. Lithium-Ion Battery Parameter Identification and State of Charge Estimation based on Equivalent Circuit Model. In Proceedings of the 2020 15th IEEE Conference on Industrial Electronics and Applications (ICIEA), Kristiansand, Norway, 9–13 November 2020.
13. Suti, A.; Di Rito, G.; Mattei, G. Heuristic Estimation of Temperature-Dependant Model Parameters of Li-Po Batteries for UAV Applications. In Proceedings of the 2023 IEEE 10th International Workshop on Metrology for AeroSpace (MetroAeroSpace), Milan, Italy, 19–21 June 2023.
14. Buddhi, W.; Izzuan, M.A.N.; Ashley, F. A multi-step parameter identification of a physico-chemical lithium-ion battery model with electrochemical impedance data. *J. Power Sources* **2023**, *580*, 233400.
15. Wu, Y.; Chen, H.; Cao, L.; Duan, J.; Chen, X.; Zhai, J. Research on Online Identification of Lithium-ion Battery Equivalent Circuit Model Parameters. In Proceedings of the 2022 9th International Forum on Electrical Engineering and Automation, Zhuhai, China, 4–6 November 2022.
16. Du, X.; Meng, J.; Liu, K.; Zhang, Y.; Wang, S.; Peng, J.; Liu, T. Online Identification of Lithium-ion Battery Model Parameters with Initial Value Uncertainty and Measurement Noise. *Chin. J. Mech. Eng.* **2023**, *36*, 7. [[CrossRef](#)]
17. Li, R.; Wang, Z.; Yu, J.; Lei, Y.; Zhang, Y.; He, J. Dynamic Parameter Identification of Mathematical Model of Lithium-Ion Battery Based on Least Square Method. In Proceedings of the 2018 IEEE International Power Electronics and Application Conference and Exposition (PEAC), Shenzhen, China, 4–7 November 2018.
18. Boukoberine, N.; Donato, T.; Benbouzid, M. Optimized Energy Management Strategy for Hybrid Fuel Cell Powered Drones in Persistent Missions Using Real Flight Test Data. *IEEE Trans. Energy Convers.* **2022**, *37*, 2080–2091. [[CrossRef](#)]
19. Di Nisio, A.; Avanzini, G.; Lotano, D.; Stigliano, D.; Lanzolla, A.M.L. Battery Testing and Discharge Model Validation for Electric Unmanned Aerial Vehicles (UAV). *Sensors* **2023**, *23*, 6937. [[CrossRef](#)] [[PubMed](#)]
20. Donato, T.; Ficarella, A.; Spedicato, L.; Arista, A.; Ferraro, M. A new approach to calculating endurance in electric flight and comparing fuel cells and batteries. *Appl. Energy* **2017**, *187*, 807–819. [[CrossRef](#)]
21. Sierra, G.; Orchard, M.; Goebel, K.; Kulkarni, C. Battery health management for small-size rotary-wing electric unmanned aerial vehicles: An efficient approach for constrained computing platforms. *Reliab. Eng. Syst. Saf.* **2019**, *182*, 166–178. [[CrossRef](#)]
22. Tang, D.Y.; Gong, M.T.; Yu, J.S.; Li, X. A power transfer model-based method for lithium-ion battery discharge time prediction of electric rotatory-wing UAV. *Microelectron. Reliab.* **2020**, *114*, 113832. [[CrossRef](#)]
23. Li, X.; Yang, Q.; Lou, Z.; Yan, W. Deep learning based module defect analysis for large scale photovoltaic farms. *IEEE Trans. Energy Convers.* **2018**, *34*, 520–529. [[CrossRef](#)]
24. Xiong, R.; Shen, W. *Advanced Battery Management Technologies for Electric Vehicles*; John Wiley & Sons, Ltd.: Hoboken, NJ, USA, 2018.
25. Feng, D.; Huang, J.; Jin, P.; Chen, H.; Wang, A.; Zheng, M. Parameter Identification and Dynamic Simulation of Lithium-Ion Power Battery Based on DP Model. In Proceedings of the 2019 14th IEEE Conference on Industrial Electronics and Applications (ICIEA), Xi'an, China, 19–21 June 2019.

26. Pai, H.-Y.; Liu, Y.-H.; Chen, G.-J.; Rasolomampionona, D.; Brescia, E.; De Tuglie, E. A novel parameters identification method of Lithium-ion battery equivalent circuit model under dynamic stress test. In Proceedings of the 2023 IEEE International Conference on Environment and Electrical Engineering and 2023 IEEE Industrial and Commercial Power Systems Europe (EEEIC/I&CPS Europe), Madrid, Spain, 6–9 June 2023.
27. He, Z.; Gao, M.; Wang, C.; Wang, L.; Liu, Y. Adaptive State of Charge Estimation for Li-Ion Batteries Based on an Unscented Kalman Filter with an Enhanced Battery Model. *J. Energy Storage* **2013**, *6*, 4134–4151. [[CrossRef](#)]
28. Shi, J.; Guo, H.; Chen, D. Parameter Identification Method for Lithium-ion Batteries Based on Recursive Least Square with Sliding Window Difference Forgetting Factor. *J. Energy Storage* **2021**, *44*, 103485. [[CrossRef](#)]
29. Li, Z.; Shi, X.; Shi, M.; Wei, C.; Di, F.; Sun, H. Investigation on the Impact of the HPPC Profile on the Battery ECM Parameters' Offline Identification. In Proceedings of the 2020 Asia Energy and Electrical Engineering Symposium (AEEES), Chengdu, China, 29–31 May 2020.
30. Huang, Y.; Li, Y.; Jiang, L.; Qiao, X.; Cao, Y.; Yu, J. Research on Fitting Strategy in HPPC Test for Li-ion battery. In Proceedings of the 2019 IEEE Sustainable Power and Energy Conference (iSPEC), Beijing, China, 21–23 November 2019.
31. Białoń, T.; Niestrój, R.; Skarka, W.; Korski, W. HPPC Test Methodology Using LFP Battery Cell Identification Tests as an Example. *Energies* **2023**, *16*, 6239. [[CrossRef](#)]
32. Frankforter, K.J.; Tejedor-Tejedor, M.I.; Anderson, M.A.; Jahns, T.M. Investigation of Hybrid Battery/Ultracapacitor Electrode Customization for Energy Storage Applications With Different Energy and Power Requirements Using HPPC Cycling. *IEEE Trans. Ind. Appl.* **2020**, *56*, 1714–1728. [[CrossRef](#)]
33. Lashway, C.R.; Mohammed, O.A. Adaptive Battery Management and Parameter Estimation Through Physics Based Modeling and Experimental Verification. *IEEE Trans. Transp. Electr.* **2016**, *2*, 454–464. [[CrossRef](#)]
34. He, H.; Zhang, X.; Xiong, R.; Xu, Y.; Guo, H. Online model-based estimation of state-of-charge and open-circuit voltage of lithium-ion batteries in electric vehicles. *Energy* **2012**, *39*, 310–318. [[CrossRef](#)]
35. Waiard, S.; Chaityut, S.; Thanatchai, K. SOC Estimation for Traction Battery Pack of Electric Vehicles with Battery Degradation Tracking. *Int. J. Intell.* **2019**, *12*, 81–89.

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